

Time: 1hr.		Class Test-2 on Computer Graphics & Geometric Modeling, 2026	Set-3	Full Marks: 30
CO1	a. Given an RGB color vector (50, 200, 150), convert it to the YIQ color model. How does adjusting the gamma value impact perceived contrast in both dark and bright regions of an image? Compare dithering and error diffusion techniques in terms of how they preserve image detail.			2+2
	b. Differentiate between Ray-casting, Ray tracing and Rediocity as techniques of rendering. Compare ambient light, diffusion reflection, and specular reflection types of illumination models.			6+3
CO2	a. Explain the role of a scene graph in organizing and managing objects within a 3D modeling environment. How can a rectangle-shaped object be efficiently inserted and updated within a linear octree structure? Discuss the method in terms of performance and spatial hierarchy.			4
	b. What is the fastest way to insert and move a rectangle object into a linear octree? How to use Ray-casting typically to implement CSG operators? Differentiate between Euclidean dimensions and Fractal dimensions?			6
CO3	a. What methods can be used to increase engagement and comfort for users in VR and AR environments? Identify non-immersive tools that can achieve similar educational or training outcomes. Why might MR offer advantages over VR for collaborative design tasks? Provide concrete examples.			7

Time: 1hr.

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Set-2

Full Marks: 30

CO1	a. A CMYK color (0.1, 0.2, 0.3, 0.1) is given. Convert it to RGB. How does the CMYK model handle colors that cannot be represented in RGB? Why might YIQ be preferred over RGB for video compression? b. What are the essential components of the 3D rendering? How reflected intensity of ambient light, diffusion reflection, and specular reflection are measured? Define depth peeling. How to determine it? Write some issues of depth peeling.	2+2 2+3+4
CO2	a. Given a Koch Curve with a segment count of 6 and a scaling factor of $1/3$, calculate its total length at the 4th iteration. Based on this, determine the fractal dimension of the curve. Describe the purpose of the firing plane in ray-casting when used to implement CSG operations. b. Explain with examples of various feasible set operations in CSG modeling. Can rendering be suitable for wireframe models? Justify your answer.	6 4
CO3	a. How does AR improve the user experience in gaming or industrial applications compared to standard 3D simulations? Describe how VR and AR can be combined to enhance training or design experiences. What are the main technical and ergonomic challenges that limit the widespread adoption of immersive technologies?	7

Time: 1hr.

Class Test-2 on Computer Graphics & Geometric Modeling, 2026

Set-1

Full Marks: 30

CO1	a. An RGB color (150, 75, 200) is given. Convert it to the HSV color space. How does changing the hue, saturation, or value affect the appearance of an image? Which applications benefit most from manipulating HSV rather than RGB? b. Define rendering and state some of its features? Distinguish between <u>online</u> and <u>offline</u> type of rendering. Compare (with examples) among flat, gouraud and phong shadings.	2+2 3+2+4
CO2	a. Compare the structures and applications of quadrees and octrees in spatial data representation. What are the limitations of using wireframe models in 3D modeling and visualization? b. Discuss how CSG operations can be applied in 3D modeling with the help of appropriate examples. Compare the usefulness of polygon meshes with CSG in representing complex solids. Also, evaluate whether wireframe models are sufficient for rendering purposes in modern graphics systems. Support your answer with valid reasoning.	4 6
CO3	a. Explain the concept of keyframe animation in computer graphics. Describe the typical workflow for generating a smooth transition between two keyframes. What roles does a 3D renderer play in multimedia applications? Discuss different storytelling metaphors (e.g., desktop, spatial, or timeline) and their impact on user interaction.	7